

Secant Varieties of Flag Varieties

Bridging Tensors, Geometry and Representation Theory

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The Problem: Tensor Rank via Secant Varieties

Motivation: Algebraic varieties can parametrize mathematical objects, such as **tensors**.

- ▶ A tensor can be viewed as a **point in projective space**.
- ▶ **Key Question:** What is the (border) rank of a tensor?

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Approach: Study **secant varieties** of varieties parametrizing rank-1 tensors.

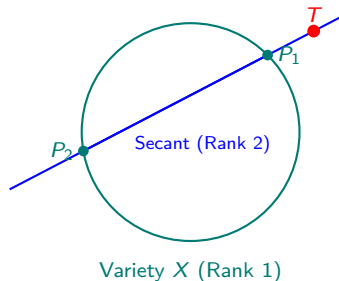
Definition: Secant Variety

Let $X \subset \mathbb{P}^N$. The s -th secant variety is:

$$\sigma_s(X) = \overline{\bigcup_{P_1, \dots, P_s \in X} \langle P_1, \dots, P_s \rangle}.$$

Classical Tool: *Terracini's Lemma* relates tangent spaces:

$$T_Q \sigma_s(X) = \langle T_{P_1} X, \dots, T_{P_s} X \rangle.$$



A solution: Tensor Rank via Interpolation

Known: Veronese Varieties

(symmetric tensors)

► **Lasker's Lemma:**

$$(\widehat{T}_P X)^\perp = I^2(p)_d.$$

- Leads to the **Dual Terracini** for Veronese varieties → **Interpolation**.

Our Result: Flag Varieties

(many structured tensors)

Dual Terracini Lemma for all flag varieties using:

- Schur apolarity
► Geometric representation theory

$$(\widehat{T}_P F)^\perp = I^{\langle 2 \rangle}(P)_\lambda.$$

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 - **Computational Framework:** Tool for computing secant dimensions of flags.
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Impact & Directions: Opens the door to systematic computational study of dimensions of secants of flag variety i.e. tensor rank of structured tensors representing complex data (biology, medicine, ...).